Seongju Lee

Ph.D. in AI Convergence

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Google Scholar



SUMMARY

Senior Researcher at HD Hyundai Robotics focused on reliable perception for robots operating in changing environments. I have led perception software development for autonomous platforms, working across dataset construction, model development, and system integration. My current work spans humanoid and dual-arm robot learning, including teleoperation and setups for collecting demonstrations and running learned policies on real robots.

SKILLS

Perception: Object Detection, Semantic Segmentation, Tracking
Learning: Domain Generalization, Knowledge Distillation, Multimodal Learning
Programming: Python, C++, C, MATLAB
Frameworks: PyTorch, ROS/ROS 2, LeRobot
Simulation: Isaac Sim, MuJoCo
Robot Learning: Imitation Learning, VLA, WAM
Robot Platforms: UR5, Franka Panda, HD Hyundai Robotics dual-arm robot
Languages: Korean (Native), English

RESEARCH EXPERIENCE

Jul. 2026 – Present — **Senior Researcher** — **HD Hyundai Robotics**
Developing intelligent dual-arm robotic systems, multimodal data pipelines, and ROS 2 interfaces.

Mar. 2026 – Jun. 2026 — **Postdoctoral Researcher** — **GIST Institute of Artificial Intelligence**

SELECTED PUBLICATIONS

Published Papers (*: Equal Contribution)

Journal metrics: 2025 JIF and best JCR subject-category quartile. Full publication list: Google Scholar.

Top Conference

MART: MultiscAle Relational Transformer Networks for Multi-agent Trajectory Prediction
The 18th European Conference on Computer Vision (ECCV 2024), Poster
TL;DR: Graph Neural Networks (GNNs), Hypergraph Transformer
Seongju Lee (First author), Junseok Lee, Yeonguk Yu, Taeri Kim, and Kyoobin Lee
[Paper] [Code] 27.9% Acceptance Rate Cited by 80

Conference

MV2: A Large-Scale 360-degree Multi-View Maritime Vision Dataset for Object Detection and Segmentation
The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)
Junseok Lee*, Jongwon Kim*, **Seongju Lee*** (Co-first author), Taeri Kim, and Kyoobin Lee
[Paper] [Website]

Top Conference

ManipForce: Force-Guided Policy Learning with Frequency-Aware Representation for Contact-Rich Manipulation
2026 IEEE International Conference on Robotics and Automation (ICRA 2026)
TL;DR: Imitation Learning, Multimodal Policy, Vision Foundation Models (VFMs), Force/Torque Sensor
Geonhyup Lee, Yeongjin Lee, Kangmin Kim, **Seongju Lee**, Sangjun Noh, Seunghyeok Back, Kyoobin Lee
[Paper] [Website] [Code] 38.04% Acceptance Rate Cited by 29

Selected Publications (continued)

Top Conference

CD-FKD: Cross-Domain Feature Knowledge Distillation for Robust Single-Domain Generalization in Object Detection

2026 IEEE International Conference on Robotics and Automation (ICRA 2026)

TL;DR: Knowledge Distillation, Robust Object Detection, Single-Domain Generalization

Junseok Lee*, Sungho Shin*, **Seongju Lee**, Kyoobin Lee

[Paper] [Code] 38.04% Acceptance Rate Cited by 1

Conference

Robust Maritime Object Detection under Adverse Conditions via Joint Semantic Learning without Extra Computational Overhead

The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)

Junseok Lee, **Seongju Lee**, Jongwon Kim, Jumi Park, and Kyoobin Lee

[Paper] 46% Acceptance Rate Cited by 2

Top Conference

Block Selection Method for Using Feature Norm in Out-of-Distribution Detection

2023 IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR 2023), Poster

Yeonguk Yu, Sungho Shin, **Seongju Lee**, Changhyun Jun, and Kyoobin Lee

[Paper] [Code] 25.8% Acceptance Rate Cited by 82

Journal

Human-efficient annotation system via active semi-supervised learning and selective review

Intelligent Systems with Applications, Vol. 31, Article 200701, 2026

Jongwon Kim, Youmin Park, Seunghyeok Hyun, Yunjae Heo, **Seongju Lee**, Yeonguk Yu, and Kyoobin Lee

[Paper] JCR 2025 Q1

Additional First-/Co-first-Author Publications

Journal

SleePyCo: Automatic Sleep Scoring with Feature Pyramid and Contrastive Learning

Expert Systems with Applications, 2024

Seongju Lee (First author), Yeonguk Yu, Seunghyeok Back, Hogeon Seo, and Kyoobin Lee

[Paper] [Code] JIF 9.4 (2025, Q1) Cited by 107

Journal

Robust Skin Disease Classification by Distilling Deep Neural Network Ensemble for the Mobile Diagnosis of Herpes Zoster

IEEE Access, 2021

Seunghyeok Back*, **Seongju Lee*** (Co-first author), Sungho Shin, Yeonguk Yu, Taekyeong Yuk, Saepomi Jong, Seungjun Ryu, Kyoobin Lee

[Paper] JIF 4.2 (2025, Q2) Cited by 80

Journal

Intra- and inter-epoch temporal context network (IITNet) using sub-epoch features for automatic sleep scoring on raw single-channel EEG

Biomedical Signal Processing and Control, 2020

Hogeon Seo*, Seunghyeok Back*, **Seongju Lee*** (Co-first author), Deokhwan Park, Tae Kim, Kyoobin Lee

[Paper] [Code] JIF 5.7 (2025, Q2) Cited by 313

Manuscript in Submission

In Submission

Rethinking Deformable Convolution as an Adapter with Cross-layer Weight Sharing for Robust Semantic Segmentation in the Wild

Target Journal: IEEE Robotics and Automation Letters

First Prize in the USV-based Obstacle Segmentation Challenge at the 3rd Maritime Computer Vision Workshop

TL;DR: Vision Foundation Models (VFMs), Domain Generalized Semantic Segmentation (DGSS), Deformable Convolution Networks (DCNs), Parameter-Efficient Fine-Tuning (PEFT)

Seongju Lee (First author), Raeyoung Kang, Sangbeom Lee, Sangjun Noh, Yeonguk Yu, Junseok Lee, and Kyoobin Lee

SELECTED PROJECTS

Jul. 2026 – Present
AI & Software Developer

Development of an Intelligent Dual-Arm Robot
HD Hyundai Robotics
TL;DR: Robot learning with VLA/WAM models; VR/glove teleoperation and hand retargeting; simulation and multimodal demonstration data pipelines.
[News]

May 2025 – Feb. 2026
AI Developer

Development of Tactile Intelligence in Robotic Hands Based on Tactile Data Learning for Manipulating Irregular Multiple Types of Objects
Ministry of Trade, Industry and Energy (MOTIE), South Korea
TL;DR: Multimodal Action Policy for Contact-Rich Tasks using Vision and F/T Sensing

Sep. 2023 – Feb. 2026
Project Member
Software Developer

Development of a Micro Intelligent Drone for Autonomous Missions in Communication/GPS-denied Environments
Institute of Civil-Military Technology Cooperation (ICMTC), South Korea
TL;DR: RGB-IR object detection and a multimodal RGB, IR, and LiDAR database captured from drone views.

Apr. 2023 – Dec. 2024
Project Member
Software Dev. Leader

Development of a Situational Awareness System for Preventing Collisions and Accidents of Autonomous Ships
Ministry of Trade, Industry and Energy (MOTIE), South Korea
TL;DR: Multi-camera maritime object detection and tracking, training dataset creation, and integration into autonomous ship systems.

Apr. 2020 – Nov. 2024
Project Leader
AI & Software Dev. Leader

Development of Swarm USV Autonomous Navigation Algorithm
Agency for Defense Development (ADD), South Korea
TL;DR: Maritime object detection and intention prediction for swarm situational awareness, integrated into multiple USVs. [News]

Mar. 2022 – Dec. 2022
Project Member
AI & Software Dev. Leader

Ceiling Layout Estimation for Indoor Environments
LG, South Korea
TL;DR: Ceiling segmentation and dataset construction for robust floor plan generation under occlusion.

EDUCATION

Mar. 2020 – Feb. 2026

Gwangju Institute of Science and Technology (GIST) **Advisor: Prof. Kyoobin Lee**
Ph.D. in AI Convergence
Dissertation Title: Robust Perception and Trajectory Forecasting for Autonomous Navigation in the Wild

Mar. 2018 – Feb. 2020

Gwangju Institute of Science and Technology (GIST)
M.S., School of Integrated Technology (Intelligent Robotics Program)

Mar. 2014 – Feb. 2018

Gwangju Institute of Science and Technology (GIST)
B.S. in Mechanical Engineering

SELECTED AWARDS

Jun. 2026

Multimodal Semantic Segmentation Challenge, 4th Workshop on Maritime Computer Vision (MaCVi) **CVPR 2026**
First Place

Jan. 2026

RoCo Challenge @ AAAI 2026: Robotic Collaborative Assembling for Human-Centered Manufacturing **AAAI, A*STAR, NTU**
Finalist Excellence Award

Nov. 2025

Autonomous Driving AI Challenge (Semantic Segmentation) **MSIT, IITP, KADIF**
Second Prize (IITP President's Award)

Dec. 2024

USV-based Obstacle Segmentation Challenge, MaCVi 2025 **WACV 2025**
First Prize

Dec. 2024

USV-based Panoptic Segmentation Challenge, MaCVi 2025 **WACV 2025**
Second Prize

Sep. 2022

The first NIA-GIST AI-HUB AI Competition **NIA, GIST**
First Prize